

Aim: Automate student marksheet generation, system information display, Fibonacci and prime number generation, and file management operations using shell scripts to enhance computational efficiency and user interaction.

- a) Write a shell script to generate mark- sheet of a student. Take 3 subjects, calculate and display total marks, percentage and Class obtained by the student.

```
user@LAPTOP-6M8BVK6P MINGW64 ~/Desktop (main)
$ echo "Enter marks of English"
read m1

echo "Enter marks of Science"
read m2

echo "Enter marks of Maths"
read m3

total=$((m1+m2+m3))
percentage=$((total/3))

echo "Student: Total Marks =$total"
echo "Percentage = $percentage"

if [ $percentage -ge 75 ];
then
    echo "Class: Distinction"
elif [ $percentage -ge 60 ];
then
    echo "Class: First Class"
elif [ $percentage -ge 50 ];
then
    echo "Class: Second Class"
elif [ $percentage -ge 35 ];
then
    echo "Class: Pass"
else
    echo "Class: Fail"
fi

Enter marks of English
70
Enter marks of Science
90
Enter marks of Maths
75
Student: Total Marks =235
Percentage = 78
Class: Distinction
```

b) Write a menu driven shell script which will print the following menu and execute the given task.

- Display calendar of current month.

```
user@LAPTOP-GMSBVK6P MINGW64 ~/Desktop (main)
$ #!/bin/bash

echo "MENU";
echo "1. Display calendar of current month";
echo "2. Display today's date and time";
echo "3. Display usernames currently logged in";
echo "4. Display your terminal number";

echo "Enter your choice:";
read choice;

if [ "$choice" -eq 1 ]; then
    echo "Calendar of current month:";
    date +"%B %Y";

elif [ "$choice" -eq 2 ]; then
    echo "Today's date and time:";
    date;

elif [ "$choice" -eq 3 ]; then
    echo "Users currently logged in:";
    whoami;

elif [ "$choice" -eq 4 ]; then
    echo "Your terminal number:";
    echo "$TERM";

else
    echo "Invalid choice";
fi
MENU
1. Display calendar of current month
2. Display today's date and time
3. Display usernames currently logged in
4. Display your terminal number
Enter your choice:
1
Calendar of current month:
January 2026
```

- Display today's date and time

```
user@LAPTOP-GMSBVK6P MINGW64 ~/Desktop (main)
$ echo "MENU";
echo "1. Display calendar of current month";
echo "2. Display today's date and time";
echo "3. Display usernames currently logged in";
echo "4. Display your terminal number";

echo "Enter your choice:";
read choice;

if [ "$choice" -eq 1 ]; then
    echo "Calendar of current month:";
    date +"%B %Y";

elif [ "$choice" -eq 2 ]; then
    echo "Today's date and time:";
    date;

elif [ "$choice" -eq 3 ]; then
    echo "Users currently logged in:";
    whoami;

elif [ "$choice" -eq 4 ]; then
    echo "Your terminal number:";
    echo "$TERM";

else
    echo "Invalid choice";
fi
MENU
1. Display calendar of current month
2. Display today's date and time
3. Display usernames currently logged in
4. Display your terminal number
Enter your choice:
2
Today's date and time:
Sat Jan 24 20:28:24 IST 2026
```

- Display usernames those are currently logged in the system.

```
user@LAPTOP-6M8BVK6P MINGW64 ~/Desktop (main)
$ echo "MENU";
echo "1. Display calendar of current month";
echo "2. Display today's date and time";
echo "3. Display usernames currently logged in";
echo "4. Display your terminal number";

echo "Enter your choice:";
read choice;

if [ "$choice" -eq 1 ]; then
    echo "Calendar of current month:";
    date +"%B %Y";

elif [ "$choice" -eq 2 ]; then
    echo "Today's date and time:";
    date;

elif [ "$choice" -eq 3 ]; then
    echo "Users currently logged in:";
    whoami;

elif [ "$choice" -eq 4 ]; then
    echo "Your terminal number:";
    echo "$TERM";

else
    echo "Invalid choice";
fi
MENU
1. Display calendar of current month
2. Display today's date and time
3. Display usernames currently logged in
4. Display your terminal number
Enter your choice:
3
Users currently logged in:
user
```

- Display your terminal number

```
user@LAPTOP-6M8BVK6P MINGW64 ~/Desktop (main)
$ #!/bin/bash

echo "MENU";
echo "1. Display calendar of current month";
echo "2. Display today's date and time";
echo "3. Display usernames currently logged in";
echo "4. Display your terminal number";

echo "Enter your choice:";
read choice;

if [ "$choice" -eq 1 ]; then
    echo "Calendar of current month:";
    date +"%B %Y";

elif [ "$choice" -eq 2 ]; then
    echo "Today's date and time:";
    date;

elif [ "$choice" -eq 3 ]; then
    echo "Users currently logged in:";
    whoami;

elif [ "$choice" -eq 4 ]; then
    echo "Your terminal number:";
    echo "$TERM";

else
    echo "Invalid choice";
fi
MENU
1. Display calendar of current month
2. Display today's date and time
3. Display usernames currently logged in
4. Display your terminal number
Enter your choice:
4
Your terminal number:
xterm
```

c) Write a shell script which will generate first n Fibonacci numbers like: 1, 1, 2, 3, 5, 13

```
user@LAPTOP-6M8BVK6P MINGW64 ~/Desktop (main)
```

```
$ #!/bin/bash
```

```
echo "Enter the value of n"
```

```
read n
```

```
a=1
```

```
b=1
```

```
echo "Fibonacci series:"
```

```
for (( i=1; i<=n; i++ ))
```

```
do
```

```
    echo -n "$a "
```

```
    c=$((a + b))
```

```
    a=$b
```

```
    b=$c
```

```
done
```

```
echo
```

```
Enter the value of n
```

```
6
```

```
Fibonacci series:
```

```
1 1 2 3 5 8
```

d) Write a shell script which will accept a number b and display first n prime numbers as output

```
user@LAPTOP-6M8BVK6P MINGW64 ~/Desktop (main)
$ #!/bin/bash
```

```
echo "Enter value of n"
read n
```

```
count=0
num=2
```

```
echo "First $n prime numbers are:"
```

```
while [ $count -lt $n ]
do
```

```
    flag=0
```

```
    for (( i=2; i<=num/2; i++ ))
    do
```

```
        if [ $((num % i)) -eq 0 ]; then
            flag=1
            break
        fi
```

```
    done
```

```
    if [ $flag -eq 0 ]; then
        echo -n "$num "
        count=$((count + 1))
    fi
```

```
    num=$((num + 1))
```

```
done
```

```
echo
```

```
Enter value of n
```

```
5
```

```
First 5 prime numbers are:
```

```
2 3 5 7 11
```

e) Write menu driven program for file handling activity

- Creation of file.

```
user@LAPTOP-6M8BVK6P MINGW64 ~/Desktop (main)
$ echo "FILE HANDLING MENU";
echo "1. Create a file";
echo "2. Write content in the file";
echo "3. Append file content";
echo "4. Delete file content";

echo "Enter your choice:";
read choice;

echo "Enter file name:";
read fname;

if [ "$choice" -eq 1 ]; then
    touch "$fname";
    echo "File created successfully";

elif [ "$choice" -eq 2 ]; then
    echo "Enter content to write (Press Ctrl+D to save):";
    cat > "$fname";
    echo "Content written successfully";

elif [ "$choice" -eq 3 ]; then
    echo "Enter content to append (Press Ctrl+D to save):";
    cat >> "$fname";
    echo "Content appended successfully";

elif [ "$choice" -eq 4 ]; then
    > "$fname";
    echo "File content deleted successfully";

else
    echo "Invalid choice";
fi
FILE HANDLING MENU
1. Create a file
2. Write content in the file
3. Append file content
4. Delete file content
Enter your choice:
1
Enter file name:
anushka
File created successfully
```

- Write content in the file.

```
user@LAPTOP-6M8BVK6P MINGW64 ~/Desktop (main)
$ echo "FILE HANDLING MENU";
echo "1. Create a file";
echo "2. Write content in the file";
echo "3. Append file content";
echo "4. Delete file content";

echo "Enter your choice:";
read choice;

echo "Enter file name:";
read fname;

if [ "$choice" -eq 1 ]; then
    touch "$fname";
    echo "File created successfully";

elif [ "$choice" -eq 2 ]; then
    echo "Enter content to write (Press Ctrl+D to save):";
    cat > "$fname";
    echo "Content written successfully";

elif [ "$choice" -eq 3 ]; then
    echo "Enter content to append (Press Ctrl+D to save):";
    cat >> "$fname";
    echo "Content appended successfully";

elif [ "$choice" -eq 4 ]; then
    > "$fname";
    echo "File content deleted successfully";

else
    echo "Invalid choice";
fi
FILE HANDLING MENU
1. Create a file
2. Write content in the file
3. Append file content
4. Delete file content
Enter your choice:
2
Enter file name:
anushka
Enter content to write (Press Ctrl+D to save):
Content written successfully
```


- Upend file content

```

user@LAPTOP-6M8BVK6P MINGW64 ~/Desktop (main)
$ echo "FILE HANDLING MENU";
echo "1. Create a file";
echo "2. Write content in the file";
echo "3. Append file content";
echo "4. Delete file content";

echo "Enter your choice:";
read choice;

echo "Enter file name:";
read fname;

if [ "$choice" -eq 1 ]; then
    touch "$fname";
    echo "File created successfully";
elif [ "$choice" -eq 2 ]; then
    echo "Enter content to write (Press Ctrl+D to save):";
    cat > "$fname";
    echo "Content written successfully";
elif [ "$choice" -eq 3 ]; then
    echo "Enter content to append (Press Ctrl+D to save):";
    cat >> "$fname";
    echo "Content appended successfully";
elif [ "$choice" -eq 4 ]; then
    > "$fname";
    echo "File content deleted successfully";
else
    echo "Invalid choice";
fi
FILE HANDLING MENU
1. Create a file
2. Write content in the file
3. Append file content
4. Delete file content
Enter your choice:
3
Enter file name:
anushka
Enter content to append (Press Ctrl+D to save):
content appended successfully

```

- Delete file content

```

user@LAPTOP-6M8BVK6P MINGW64 ~/Desktop (main)
$ echo "FILE HANDLING MENU";
echo "1. Create a file";
echo "2. Write content in the file";
echo "3. Append file content";
echo "4. Delete file content";

echo "Enter your choice:";
read choice;

echo "Enter file name:";
read fname;

if [ "$choice" -eq 1 ]; then
    touch "$fname";
    echo "File created successfully";
elif [ "$choice" -eq 2 ]; then
    echo "Enter content to write (Press Ctrl+D to save):";
    cat > "$fname";
    echo "Content written successfully";
elif [ "$choice" -eq 3 ]; then
    echo "Enter content to append (Press Ctrl+D to save):";
    cat >> "$fname";
    echo "Content appended successfully";
elif [ "$choice" -eq 4 ]; then
    > "$fname";
    echo "File content deleted successfully";
else
    echo "Invalid choice";
fi
FILE HANDLING MENU
1. Create a file
2. Write content in the file
3. Append file content
4. Delete file content
Enter your choice:
4
Enter file name:
anushka
File content deleted successfully

```